# Week 3: Estimation of Discrete Choice Models

## Logit, GEV, Probit, and Mixed Logit

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# Roadmap

1. Random Utility Framework
2. Multinomial Logit
   - Choice probabilities
   - Maximum likelihood
   - Identification and normalization
   - Marginal effects and interpretation
   - IIA and substitution patterns
3. Generalized Extreme Value Models
   - Nested Logit
   - Inclusive value
   - Correlated alternatives
4. Multinomial Probit
5. Mixed Logit
6. Unified comparison

# 1. Random Utility Framework

Individual $i$ chooses one alternative from the choice set

$$\mathcal{J} = \{1, \ldots, J\}.$$

Utility from choosing alternative $j$ is

$$U_{ij} = V_{ij} + \varepsilon_{ij}$$

where

$$V_{ij} = x_{ij}'\beta$$

is the systematic component of utility, and

$$\varepsilon_{ij}$$

captures utility components observed by the decision maker but not by the econometrician.

The observed choice satisfies

$$y_i = j \iff U_{ij} \geq U_{ik}, \qquad \forall k \neq j.$$

# Choice Probability

Since $\varepsilon_{ij}$ is unobserved, choice is probabilistic from the econometrician's perspective.

$$P_{ij} = \Pr(y_i = j)$$

or equivalently,

$$P_{ij} = \Pr\left(U_{ij} > U_{ik}, \forall k \neq j\right).$$

Substituting the random utility representation,

$$P_{ij} = \Pr\left(\varepsilon_{ik} - \varepsilon_{ij} < V_{ij} - V_{ik}, \forall k \neq j\right).$$

## Central econometric question

The distribution assumed for

$$\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iJ})$$

determines the choice probabilities and therefore the discrete choice model.

# The Overall Steps of Discrete Choice Estimation

The entire class of models can be organized as

$$\text{Utility specification} \rightarrow \text{Distribution of } \varepsilon \rightarrow \text{Choice probabilities}$$

$$\rightarrow \text{Likelihood} \rightarrow \text{Estimation}$$

Different assumptions about unobserved utility imply different:

- substitution patterns;
- correlations across alternatives;
- computational requirements;
- identification restrictions.

# 2. Multinomial Logit

The Multinomial Logit model assumes

$$\varepsilon_{ij} \overset{iid}{\sim} \text{Type-I Extreme Value}$$

with cumulative distribution function

$$F(\varepsilon) = \exp\{-\exp(-\varepsilon)\}.$$

The key assumptions are:

- identical marginal distributions;
- independence across alternatives;
- independence across decision makers;
- a fixed scale normalization.

The major payoff is a closed-form choice probability.

# Logit Choice Probability

Under i.i.d. Type-I Extreme Value errors,

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_{k=1}^{J} \exp(V_{ik})}$$

and, with

$$V_{ij} = x_{ij}'\beta,$$

we obtain

$$P_{ij}(\beta) = \frac{\exp(x_{ij}'\beta)}{\sum_{k=1}^{J} \exp(x_{ik}'\beta)}.$$

This closed form is the main computational advantage of the Logit model.

# From Choice Probabilities to Likelihood

Define the indicator

$$y_{ij} = \begin{cases} 1, & \text{if individual } i \text{ chooses } j, \\ 0, & \text{otherwise.} \end{cases}$$

Individual $i$ contributes

$$L_i(\beta) = \prod_{j=1}^{J} P_{ij}(\beta)^{y_{ij}}.$$

Assuming independent observations,

$$L(\beta) = \prod_{i=1}^{N} \prod_{j=1}^{J} P_{ij}(\beta)^{y_{ij}}.$$

Taking logs,

$$\ell(\beta) = \sum_{i=1}^{N} \sum_{j=1}^{J} y_{ij} \log P_{ij}(\beta).$$

# Maximum Likelihood Estimation

The estimator is

$$\hat{\beta} = \arg\max_{\beta} \ell(\beta).$$

For Logit,

$$\log P_{ij} = V_{ij} - \log\left(\sum_k e^{V_{ik}}\right).$$

Therefore,

$$\ell(\beta) = \sum_i \sum_j y_{ij}\left[x_{ij}'\beta - \log\left(\sum_k e^{x_{ik}'\beta}\right)\right].$$

Estimation therefore follows the sequence

$$U_{ij} \to P_{ij}(\beta) \to \ell(\beta) \to \hat{\beta}.$$

# Score and First-Order Conditions

For the standard Logit specification,

$$\frac{\partial \ell(\beta)}{\partial \beta} = \sum_i \sum_j (y_{ij} - P_{ij})\, x_{ij}.$$

The first-order condition is

$$\sum_i \sum_j \left( y_{ij} - P_{ij}(\hat{\beta}) \right) x_{ij} = 0.$$

Thus maximum likelihood chooses $\hat{\beta}$ such that predicted choices systematically match observed choices with respect to the included covariates.

Under standard regularity conditions,

$$\sqrt{N}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \mathcal{I}^{-1}),$$

where $\mathcal{I}$ is the Fisher information matrix.

# 3. Identification: Only Utility Differences Matter

Choice depends on utility rankings, not absolute utility levels.
For two alternatives,

$$U_{i1} > U_{i0}$$

is equivalent to

$$U_{i1} - U_{i0} > 0.$$

Therefore, adding the same constant $c_i$ to all alternatives does not change behavior:

$$U_{ij}^{*} = U_{ij} + c_i.$$

Hence absolute utility levels are not identified.

A common normalization is to choose a reference alternative:

$$V_{i0} = 0.$$

Only utility differences relative to the reference alternative are identified.

# Scale Normalization

Choice rankings are also invariant to multiplication by a positive constant:

$$U_{ij}^{*} = cU_{ij}, \qquad c > 0.$$

Therefore,

$$U_{ij} > U_{ik} \iff cU_{ij} > cU_{ik}.$$

The scale of utility is not identified separately from the variance of the error term.

More precisely, choice data identify

$$\frac{\beta}{\sigma_{\varepsilon}},$$

rather than $\beta$ and $\sigma_{\varepsilon}$ separately.

## Implication

Coefficient magnitudes from different Logit models or samples cannot always be directly compared because their underlying utility scales may differ.

# Alternative-Specific and Individual-Specific Variables

A typical utility specification is

$$V_{ij} = \alpha_j + x_{ij}'\beta + z_i'\gamma_j.$$

Three types of variables may appear:

1. Alternative-specific characteristics

   $$x_{ij}$$

   such as price, travel time, distance.
2. Individual characteristics

   $$z_i$$

   such as income or age.
3. Alternative-specific constants

   $$\alpha_j.$$

Individual-specific variables alone do not vary across alternatives and therefore usually enter through interactions with alternatives.

# 4. Interpretation: Coefficients Are Utility Parameters

Suppose

$$V_{ij} = \beta_p p_{ij} + \beta_t t_{ij}.$$

Then

$$\beta_p$$

measures how price affects latent utility, not directly how price affects choice probability. Since probabilities are nonlinear,

$$\frac{\partial P_{ij}}{\partial x} \neq \beta.$$

For an alternative-specific variable $x_{ij}$,

$$\frac{\partial P_{ij}}{\partial x_{ij}} = \beta P_{ij}(1 - P_{ij}),$$

while for $k \neq j$,

$$\frac{\partial P_{ik}}{\partial x_{ij}} = -\beta P_{ij} P_{ik}$$

# Willingness to Pay

Suppose utility contains an environmental amenity $q_{ij}$ and price $p_{ij}$:

$$V_{ij} = \beta_q q_{ij} + \beta_p p_{ij} + \cdots$$

with

$$\beta_p < 0.$$

The marginal willingness to pay for the amenity is

$$MWTP = -\frac{\beta_q}{\beta_p}.$$

An advantage of coefficient ratios is that the common utility scale cancels:

$$-\frac{\beta_q/\sigma}{\beta_p/\sigma} = -\frac{\beta_q}{\beta_p}.$$

This explains why willingness-to-pay measures are central in many applied discrete choice models.

# 5. IIA: The Central Restriction of Logit

Consider two alternatives $j$ and $k$.
Under Logit,

$$P_{ij} = \frac{e^{V_{ij}}}{\sum_l e^{V_{il}}}, \qquad P_{ik} = \frac{e^{V_{ik}}}{\sum_l e^{V_{il}}}.$$

Therefore,

$$\frac{P_{ij}}{P_{ik}} = \exp(V_{ij} - V_{ik}).$$

The relative odds of choosing $j$ rather than $k$ are independent of the characteristics or existence of any other alternatives.

This is the

Independence of Irrelevant Alternatives (IIA).

# The Red Bus / Blue Bus Problem

Suppose initially the choice set is

$$\{\text{Car}, \text{Bus}\}$$

with

$$P(\text{Car}) = 0.5, \qquad P(\text{Bus}) = 0.5.$$

Now divide Bus into two nearly identical alternatives:

$$\{\text{Car}, \text{Red Bus}, \text{Blue Bus}\}.$$

If Red Bus and Blue Bus are essentially identical, economic intuition suggests that the new alternative should primarily take market share from the original Bus option.

Standard Logit instead preserves relative odds according to IIA.

# Where Does IIA Come From?

The problem does not originate from maximum likelihood estimation.
It originates from the stochastic utility assumption

$$\varepsilon_{ij} \perp \varepsilon_{ik}.$$

If Red Bus and Blue Bus share unobserved characteristics, one would expect

$$\mathrm{Cov}(\varepsilon_{i,\mathrm{Red}}, \varepsilon_{i,\mathrm{Blue}}) > 0.$$

But standard Logit imposes

$$\mathrm{Cov}(\varepsilon_{ij}, \varepsilon_{ik}) = 0, \qquad j \neq k.$$

Thus the natural extension is to introduce correlation in unobserved utility.

$$\text{Logit} \rightarrow \text{correlated alternatives} \rightarrow \text{GEV models.}$$

# 6. Generalized Extreme Value Models

The Generalized Extreme Value (GEV) family relaxes the independence structure of the standard Logit model while preserving analytically tractable choice probabilities.

If

$$y_j = e^{V_j},$$

a GEV model can be characterized by a generating function

$$G(y_1, \ldots, y_J).$$

Choice probabilities take the form

$$P_j = \frac{y_j G_j(y_1, \ldots, y_J)}{G(y_1, \ldots, y_J)},$$

where

$$G_j = \frac{\partial G}{\partial y_j}.$$

The Multinomial Logit is a special case of the GEV family.

# Why GEV?

GEV models are designed to balance two objectives:

1. Allow economically meaningful correlation among alternatives;
2. Preserve closed-form or highly tractable choice probabilities.

The most widely used example is

Nested Logit.

Its central idea is simple:

*Alternatives that are similar to each other can share a common unobserved component.*

# 7. Nested Logit: Example

Suppose travelers choose among

$$\{\text{Car}, \text{Bike}, \text{Bus}, \text{Metro}\}.$$

A plausible nesting structure is

$$\begin{aligned} \text{Private} &: \{\text{Car}, \text{Bike}\}, \\ \text{Public Transit} &: \{\text{Bus}, \text{Metro}\}. \end{aligned}$$

Bus and Metro may share unobserved characteristics such as

- public transport accessibility;
- inconvenience from transfers;
- exposure to congestion;
- preferences toward collective transportation.

Nested Logit allows their unobserved utilities to be correlated.

# Probability Decomposition

Suppose alternative $j$ belongs to nest $m$.
Then

$$P_{ij} = P_{ij|m} P_{im}.$$

That is,

$$\Pr(i \text{ chooses } j) = \Pr(i \text{ chooses nest } m) \times \Pr(i \text{ chooses } j \mid m).$$

For example,

$$P_i(\text{Bus}) = P_i(\text{Public Transit}) \times P_i(\text{Bus} \mid \text{Public Transit}).$$

The model therefore separates

within-nest substitution

from

between-nest substitution.

# Conditional Choice Within a Nest

For alternative $j$ in nest $m$,

$$P_{ij|m} = \frac{\exp(V_{ij}/\lambda_m)}{\sum_{k \in m} \exp(V_{ik}/\lambda_m)}.$$

The parameter

$$\lambda_m$$

is the nest-specific dissimilarity parameter.

Define the inclusive value

$$IV_{im} = \log \left[ \sum_{j \in m} \exp(V_{ij}/\lambda_m) \right].$$

This summarizes the attractiveness of all alternatives inside nest $m$.

# The Inclusive Value

The inclusive value

$$IV_{im} = \log \sum_{j \in m} e^{V_{ij}/\lambda_m}$$

is also called the

log-sum.

It represents the expected maximum utility available from alternatives within the nest, up to normalization.

Suppose Metro becomes faster.
Then

$$V_{i,\text{Metro}} \uparrow$$

implies

$$IV_{i,\text{Public Transit}} \uparrow .$$

Thus improving one alternative makes the entire nest more attractive.

# Probability of Choosing the Nest

Allow the upper-level utility of nest $m$ to contain observed characteristics $W_{im}$.
Then

$$P_{im} = \frac{\exp\left(W_{im} + \lambda_m IV_{im}\right)}{\sum_n \exp\left(W_{in} + \lambda_n IV_{in}\right)}.$$

Combining both levels,

$$P_{ij} = P_{im} P_{ij|m}.$$

Hence Nested Logit remains analytically tractable and can be estimated by maximum likelihood.

# Interpretation of the Dissimilarity Parameter

The parameter

$$\lambda_m$$

controls substitution within nest $m$.
Under the standard Random Utility Model consistency conditions,

$$0 < \lambda_m \leq 1.$$

A useful benchmark is

$$\lambda_m = 1.$$

When all nest parameters equal one, the nesting structure disappears and the model collapses to standard Multinomial Logit.

Smaller $\lambda_m$ implies stronger similarity among alternatives within the nest under the usual parameterization.

# Logit as a Special Case

Nested Logit therefore provides a direct extension of Logit:

$$\text{Multinomial Logit} \subset \text{GEV family}.$$

The conceptual progression is

$$\text{i.i.d. EV1 errors}$$

$$\Downarrow$$

$$\text{IIA across all alternatives}$$

$$\Downarrow$$

$$\text{allow structured correlation}$$

$$\Downarrow$$

# Estimation of Nested Logit

Since the unconditional probability is

$$P_{ij}(\theta) = P_{im}(\theta) P_{ij|m}(\theta),$$

the log-likelihood remains

$$\ell(\theta) = \sum_i \sum_j y_{ij} \log P_{ij}(\theta).$$

The parameter vector may contain

$$\theta = (\beta, \gamma, \lambda_1, \ldots, \lambda_M).$$

Two approaches are possible:

- Sequential estimation;
- Full-information maximum likelihood.

In modern applications, joint maximum likelihood estimation is generally preferred when computationally feasible.

# What Does Nested Logit Solve?

Nested Logit solves an important limitation of standard Logit:

all alternatives need not be equally substitutable.

It permits stronger substitution among alternatives in the same nest.

But it does **not** generate completely unrestricted substitution patterns.
Results depend on the specified nesting structure.

## Economic requirement

Nests should represent economically meaningful patterns of similarity, rather than being selected only to improve statistical fit.

# 8. Multinomial Probit

An alternative way to relax Logit's independence assumption is to allow correlation in the latent utility shocks:

$$\varepsilon_i \sim N(\mathbf{0}, \Sigma).$$

The covariance matrix allows flexible correlation among alternatives:
Therefore,

$$\text{Cov}(\varepsilon_{ij}, \varepsilon_{ik}) \neq 0$$

can be estimated, subject to the location and scale normalizations required because only utility differences determine choice.

This produces much more flexible substitution patterns than standard Logit.

# The Cost of Flexibility in Probit

For alternative $j$, define the utility-difference vector relative to $j$:

$$\boldsymbol{d}_{ij} = (U_{ik} - U_{ij})_{k \neq j}, \qquad \boldsymbol{d}_{ij} \sim \mathcal{N}(\boldsymbol{\mu}_{ij}, \Omega_j),$$

where $\mu_{ij,k} = V_{ik} - V_{ij}$ and

$$\Omega_j = \mathrm{Var}\left((\varepsilon_{ik} - \varepsilon_{ij})_{k \neq j}\right).$$

The choice probability is the probability that every component is negative:

$$\begin{aligned} P_{ij} &= \Pr(\boldsymbol{d}_{ij} < \mathbf{0} \mid X_i) \\ &= \int_{(-\infty, 0]^{J_i - 1}} \phi_{J_i - 1}(\boldsymbol{d}; \boldsymbol{\mu}_{ij}, \Omega_j)\, d\boldsymbol{d}. \end{aligned}$$

Under multivariate normal errors, this generally requires evaluating a multivariate normal integral. There is generally no convenient closed-form expression when $J_i$ is larger than two.

Hence estimation may require

- numerical integration;
- simulation;
- specialized algorithms for multivariate normal probabilities.

The basic trade-off is therefore

# 9. Mixed Logit: Preference Heterogeneity

Previous models primarily changed the structure of $\varepsilon$.
Mixed Logit instead allows preference parameters themselves to vary across individuals.

$$\beta_i = \bar{\beta} + \eta_i.$$

For example,

$$\beta_{p,i} \sim N(\bar{\beta}_p, \sigma_p^2).$$

Different individuals can therefore have different sensitivities to

- price;
- travel time;
- environmental quality;
- product characteristics.

# Conditional Logit Representation

Conditional on a particular value of $\beta$, choice remains Logit:

$$L_{ij}(\beta) = \frac{\exp(x_{ij}'\beta)}{\sum_k \exp(x_{ik}'\beta)}.$$

But $\beta_i$ is unobserved.
Suppose

$$\beta_i \sim f(\beta \mid \theta).$$

Then the unconditional probability is

$$P_{ij} = \int L_{ij}(\beta) f(\beta \mid \theta)\, d\beta.$$

Thus Mixed Logit is literally a mixture of Logit probabilities over the distribution of heterogeneous preferences.

# Simulated Maximum Likelihood

The integral

$$P_{ij} = \int L_{ij}(\beta) f(\beta \mid \theta) d\beta$$

usually has no closed-form solution.
Draw

$$\beta_i^{(1)}, \ldots, \beta_i^{(R)} \sim f(\beta \mid \theta).$$

Approximate the probability by

$$\hat{P}_{ij} = \frac{1}{R} \sum_{r=1}^{R} L_{ij}\left(\beta_i^{(r)}\right).$$

Then maximize the simulated log-likelihood

$$\hat{\ell}(\theta) = \sum_i \sum_j y_{ij} \log \hat{P}_{ij}(\theta).$$

This is Simulated Maximum Likelihood.

# 10. A Unified View

Start from

$$U_{ij} = x_{ij}'\beta_i + \varepsilon_{ij}.$$

Then ask two questions.

**Question 1: How are the errors distributed?**

$$\varepsilon_{ij} \overset{iid}{\sim} EV1 \qquad \Rightarrow \qquad \text{Logit}$$

$$\varepsilon_{ij} \text{ correlated according to a GEV structure} \qquad \Rightarrow \qquad \text{GEV / Nested Logit}$$

$$\varepsilon_i \sim N(0, \Sigma) \qquad \Rightarrow \qquad \text{Probit.}$$

# A Unified View: Preference Heterogeneity

**Question 2: Is $\beta$ the same for everyone?**
Standard Logit, Nested Logit, and basic Probit commonly begin with

$$\beta_i = \beta.$$

Mixed Logit instead assumes

$$\beta_i \sim f(\beta \mid \theta).$$

Hence two major sources of flexibility are

$$\text{correlation in } \varepsilon$$

and

$$\text{heterogeneity in } \beta.$$

These are conceptually distinct mechanisms.

# Comparison of the Four Models

| Model | Main Assumption | Substitution | Probability | Estimation |
|---|---|---|---|---|
| Logit | i.i.d. EV1 errors | IIA | Closed form | MLE |
| GEV / Nested Logit | Structured correlation in EV errors | Flexible within predefined structure | Closed form | MLE |
| Probit | Multivariate normal errors | Flexible covariance structure | Multivariate integral | Numerical / simulation |
| Mixed Logit | Random coefficients | Highly flexible | Integral over coefficients | Simulated MLE |

# What Changes Across Models?

The four models should not be viewed as four unrelated probability formulas.
They answer different versions of the same question:

What structure should we impose on unobserved utility?

Logit : independent EV1 shocks

GEV : structured correlation

Probit : general normal covariance

Mixed Logit : random preference parameters.

Each assumption generates a different economic model of substitution.

# 11. Main Takeaways

1. Discrete choice begins with the Random Utility Model:

$$U_{ij} = V_{ij} + \varepsilon_{ij}.$$

2. Distributional assumptions on $\varepsilon$ determine choice probabilities.
3. Logit provides closed-form probabilities and simple MLE.
4. Utility requires both location and scale normalization.
5. Logit's i.i.d. error assumption implies IIA.
6. GEV / Nested Logit relaxes IIA by introducing structured correlation.
7. Probit allows flexible error correlation but sacrifices closed-form probabilities.
8. Mixed Logit introduces continuous preference heterogeneity and typically requires simulation.

# The Core Logic

The key sequence to remember is

Distributional assumptions

$\Downarrow$

Choice probabilities

$\Downarrow$

Substitution patterns

$\Downarrow$

Likelihood

$\Downarrow$

# From Discrete Choice to Structural Demand

The same logic provides the foundation for more advanced models:

Logit

$\Downarrow$

Random-Coefficient Logit

$\Downarrow$

Aggregate Demand Estimation

$\Downarrow$

BLP-type models.

It also provides the static foundation for

$$U_{ij} = x'_{ij}\beta_i + \varepsilon_{ij}$$

Understanding discrete choice estimation means understanding how assumptions about $\beta_i$ and $\varepsilon_{ij}$ translate into observable choice behavior.

**End**

# Running example: residential sorting

**Bayer, Ferreira, and McMillan (2007), JPE**

*A Unified Framework for Measuring Preferences for Schools and Neighborhoods*

- Households choose among residential locations that bundle housing, schools, neighborhood composition, and access.
- Observed choices reveal preferences through comparisons across feasible alternatives.
- The paper combines a heterogeneous residential choice model with a boundary discontinuity design.

## Teaching value

The application connects utility, choice probabilities, heterogeneity, endogeneity, identification, and willingness to pay.

# Residential utility

$$U_{ij} = \alpha_i p_j + \beta_i S_j + \gamma_i N_j + \delta_i A_{ij} + \xi_j + \varepsilon_{ij}.$$

- $p_j$: housing price or expenditure.
- $S_j$: school quality.
- $N_j$: neighborhood attributes.
- $A_{ij}$: access and distance variables.
- $\xi_j$: unobserved location quality shared across households.
- $\varepsilon_{ij}$: household-specific idiosyncratic taste.

Household characteristics can shift the coefficients:

$$(\alpha_i, \beta_i, \gamma_i, \delta_i) = Bz_i + \eta_i.$$

# From utility to residential choice

$$y_i = j \quad \Longleftrightarrow \quad U_{ij} > U_{ik}, \quad \forall k \neq j.$$

Under a conditional logit benchmark,

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_{k \in \mathcal{J}_i} \exp(V_{ik})},$$

where

$$V_{ij} = \alpha_i p_j + \beta_i S_j + \gamma_i N_j + \delta_i A_{ij} + \xi_j.$$

$$L_i = \prod_{j \in \mathcal{J}_i} P_{ij}^{1\{y_i = j\}}.$$

## Interpretation

The school coefficient becomes a willingness-to-pay parameter only after the price coefficient and the source of identifying variation are credible.

# Heterogeneity and residential sorting

$$\beta_i = \beta_0 + \beta_1 \,\mathrm{Income}_i + \eta_i, \qquad \gamma_i = \gamma_0 + \gamma_1 \,\mathrm{Children}_i + \nu_i.$$

- Households may value school quality differently.
- Income can change the marginal utility of expenditure and school quality.
- Families with children may sort on school attributes more strongly.
- Sorting changes neighborhood composition, so raw correlations do not directly identify causal preferences.

Mixed logit integrates over $f(\boldsymbol{\beta}_i \mid z_i, \boldsymbol{\theta})$ rather than imposing one common slope vector.

# The endogeneity problem

Suppose unobserved neighborhood quality enters utility:

$$U_{ij} = \cdots + \beta S_j + \xi_j + \varepsilon_{ij}.$$

If

$$\mathrm{Cov}(S_j, \xi_j) \neq 0,$$

then a naive estimate of $\beta$ confounds school preferences with other location attributes.

- Desirable neighborhoods may have both better schools and better unobserved amenities.
- Higher-income households may sort into multiple desirable attributes simultaneously.
- Housing prices capitalize expected neighborhood quality.

The identification question is: what variation in school quality changes utility while holding unobserved location quality sufficiently fixed?

# Boundary discontinuities

School attendance boundaries can generate sharp changes in assigned school quality across nearby locations.

- Compare households living close to opposite sides of a boundary.
- Nearby homes can be similar while school assignment changes discontinuously.
- The design reduces confounding from broad neighborhood differences.

Conceptually,

$$S(r) = S_{-} + \Delta S 1\{r \geq 0\} + u(r),$$

where $\Delta S$ is the discontinuous change at the boundary.

## Key identifying assumption

Absent the boundary-induced assignment change, relevant outcomes would vary smoothly through the boundary.

# Identification in the structural model

The boundary design is informative when:

1. households cannot precisely manipulate location in response to an unobserved discontinuity;
2. other housing and neighborhood attributes change smoothly at the boundary;
3. the choice set and assignment rule are correctly represented;
4. the model separates observed attributes from persistent location quality.

The local quasi-experiment helps identify the causal school-quality parameter, while the broader choice data identify heterogeneous preferences and substitution across locations.

$$\text{local design} + \text{global choice model} \quad \Rightarrow \quad \text{interpretable preference estimates.}$$

# Willingness to pay

For a marginal change in school quality offset by a price change,

$$dU_i = \beta_i \, dS + \alpha_i \, dp = 0.$$

Therefore,

$$\text{MWTP}_{i,S} = -\frac{\beta_i}{\alpha_i}.$$

- If $\alpha_i < 0$ and $\beta_i > 0$, willingness to pay is positive.
- Heterogeneous coefficients imply a distribution of willingness to pay.
- Scale normalization cancels in the ratio under a common utility scale.
- Counterfactual welfare requires the full choice model and the relevant alternative set.

# Final synthesis

1. Random utility turns preference comparisons into choice probabilities.
2. Error assumptions determine substitution patterns and computational requirements.
3. Identification requires normalization, useful variation, and credible treatment of unobserved attributes.
4. Logit is transparent; GEV adds structured dependence while retaining closed-form probabilities.
5. Probit and mixed logit allow greater flexibility at the cost of integration and simulation.
6. The residential sorting application shows why structural modeling and research design must be considered together.

# Reference

Bayer, P., F. Ferreira, and R. McMillan. 2007. "A Unified Framework for Measuring Preferences for Schools and Neighborhoods." *Journal of Political Economy* 115(4): 588–638.

`https://www.nber.org/papers/w13236`